

Md. Masud Mazumder

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Education

University of Manitoba

M.Sc. in Computer Science — GPA: 4.38/4.50 (coursework completed)
Supervisor: Dr. Shaiful Chowdhury — SQM Research Lab

Winnipeg, MB, Canada

September 2025 – Present

- **Selected Thesis Topic:** Software vulnerability and property-based testing
- **Relevant Coursework:** Deep Generative Modeling: Theory and Application (A+) and Software Quality (A+)

University of Chittagong

B.Sc. in Computer Science and Engineering — CGPA: 3.84/4.00

Chittagong, Bangladesh

January 2019 – October 2024

- Ranked **2nd** out of 82 students
- **Thesis:** *PairSiamNet: Transformer-Enhanced Siamese Network with Pairwise Learning for Semantic Relatedness* — Evaluated semantic relatedness between sentence pairs using pairwise contrastive learning

Publications

Method-Level Bug Prediction: How Far Are We?

- Md Nahidul Islam Opu, Md. Masud Mazumder, Muhammad Asaduzzaman, Shaiful Chowdhury - ICSE 2026 (Research Track) - **Under review**
- Developed a realistic method-level bug prediction using LLM-based label cleaning, code embeddings, and time-sensitive cross-project evaluation, revealing persistent generalization challenges.

How Do Agentic AI Systems Deal With Software Energy Concerns? A Pull Request-Based Study

- Tanjum Motin Mitul, Md. Masud Mazumder, Shaiful Chowdhury (*Equal contribution with the first author*) - MSR 2025 (Challenge Track) - Paper Link
- Applied thematic analysis to 216 energy-explicit pull requests in LLM-based coding agents; produced a taxonomy of five energy concern categories and 21 optimization techniques, and analyzed PR acceptance patterns across the AIDev dataset.

Work Experiences

Lecturer

Dept. of Computer Science and Engineering, East Delta University

Chittagong, Bangladesh

January 2025 – July 2025

- Taught undergraduate computer science courses and designed assessments

Technical Skills

AI / ML: PyTorch, Tensorflow, LangGraph & LangChain, MCP & Tools

Languages: C, C++, Python, Java, JavaScript, PHP

Web & Mobile: ReactJS, Next.js, Express.js, Flutter, React Native, HTML, CSS

Databases: MySQL, PostgreSQL, MongoDB

Tools: Git & GitHub, L^AT_EX

Completed Projects

- Orion** | *TypeScript, LangGraph & langChain, ElectronJS, ReactJS, SQLite, Ollama* | [GitHub Link](#) 2026
- This personal project aims to build a secure local assistant that executes agentic workflows for both text and voice commands. The ElectronJS- and ReactJS-based desktop application provides the user interface, while locally or remotely hosted LLMs power the custom agentic workflow.
- Quality Per Joule** | *Python, PyTorch, CodeCarbon, Generative Models* | [GitHub Link](#) 2026
- Course project for Deep Generative Modeling in which image quality gain per joule was measured across several generative model families, including AE, VAE, GAN, WGAN, and DDPM. According to the result, AE outperforms when both image quality and energy consumption are considered.
- EcoSync** | *ReactJS, Node.js, PostgreSQL, React Native* | [GitHub Link](#) 2024
- Full-stack waste management platform for Dhaka North City Corporation with real-time vehicle tracking, an admin dashboard, user authentication, and a companion mobile app for community interaction.
- qCompiler** | *Node.js* | [GitHub Link](#) 2024
- Open-source Node.js API library for building online and offline code compiler applications; supports multiple languages with a unified interface.
- CUOPAS — Student Management System** | *ReactJS, Node.js/Express, MySQL* | [GitHub Link](#) 2023
- Comprehensive student management platform covering enrollment, attendance tracking, examination management, and automated result processing; awarded **3rd Place** at the Engineering Day Project Presentation, University of Chittagong.

Competitions & Participations

- Safe Food For Canada Hackathon** | University of Manitoba & University of Guelph 2025
- **2nd Place** (team) among participating universities
- IEEE Xtreme Programming Competition 19.0** 2025
- Ranked **102nd** in Canada
- Inter-University Hackathon — Code Samurai 2024** | University of Dhaka & BJIT Limited 2024
- **Finalist** among 450+ teams from universities across Bangladesh
- Engineering Day Programming Contest** | University of Chittagong 2023
- **2nd Place** this programming contest focused on algorithmic soundness and mathematical analysis
- Engineering Day Project Presentation** | University of Chittagong 2023
- **3rd Place** (team) — Presented a student management system covering enrollment, attendance, examination, and result processing (ReactJS, Node.js/Express.js, MySQL)
- Inter-University Programming Contest (Divisional)** | CUET, Chittagong 2022
- **5th Place** (team) among divisional inter-university participants

Certifications

- Reinforcement Learning Specialization** | Coursera, University of Alberta *In Progress*
- Instructors: Prof. Adam White and Prof. Martha White, University of Alberta
- Fundamental Information Technology Engineer Examination** April 2023
- Bangladesh IT-Engineers Examination Center, Bangladesh Computer Council (ranked 7th nationwide)